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Strong-Lensing Time Delays in the Berger–Hopf Framework: An R1 Effective Calculation

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Abstract

In the Berger–Hopf interpretation, the optical transfer of prepared environmental light depends on the envelope state. We construct a reproducible strong-lensing benchmark from a specified R1 cubic scalar–tensor action and ask which properties of that transfer ordinary lens models can identify. A common incident marker, physical source, observer clock, redshifts and distance maps define the comparison. The regular source determines the scalar and metric response without a fitted local amplitude. For the declared $10^{11} M_{\odot}$, 3 kpc body, its screening radius is 212.0292 kpc and its fixed-body-reference delay offsets are 25.9593 s and 8.6490 s. A positive conventional effective mass reproduces the complete retained response. A finite five-parameter conventional refit also reproduces the three injected images and two delays to numerical precision. A smooth profile matched by mass and half-mass radius changes the offsets while preserving this equivalence. State-coefficient derivatives and common-gravity and distributed-encounter controls quantify the transfer's dependencies and approximation hierarchy. The result is a conditional forward and inverse benchmark, with a microscopic BHSM source embedding and evolved outer state identified separately.

Keywords: Berger–Hopf framework; strong gravitational lensing; time delays; envelope response; Vainshtein screening; effective-mass representation

1. Introduction

In the Berger–Hopf Standard Model (BHSM) interpretation, gravitational lensing is the optical response of an envelope event traversed by environmental radiation. The BHSM program asks how envelope-state and surface contributions determine that response. Prepared incoming light is related to outgoing image directions and observer arrival-time differences, conditional on the envelope state and surrounding geometry. This raises a modeling question: which properties of a state-conditioned transfer can be identified once ordinary lens and source freedoms are admitted?

We address that question through one explicit conditional realization. The previously specified closed reference branch R1 supplies a cubic scalar–tensor action and frozen rolling coefficients [1]. Its regular finite-body source determines the metric and projected encounter without a fitted local amplitude. The same emitted marker, physical source coordinate, measured redshifts, observer clock and compatible distance maps [2] are used for the response and its conventional fixed-body reference. This common preparation makes their difference a defined forward calculation.

Cubic kinetic braiding, screening and their effects on lensing are established mechanisms [3–5]. Lensing inference also depends on the assumed profile, mass sheet, source

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and propagation geometry [6–9]. Our contribution is a controlled finite-body forward benchmark together with an explicit test of what its image and delay response identifies. It combines the normalized scalar source, one consistent potential/gradient/Hessian, resolved image branches, a positive conventional effective-mass representation and a finite nonlinear nuisance refit. Smooth-profile and distributed-encounter controls test whether the conclusions depend on the convenient body or optical plane. This is the practical role of a bounded forward model with stated limitations [10].

The calculation follows state and preparation, normalized source and boundary problem, metric/optical transfer, images and delays, then identifiability and approximation controls. A positive effective mass reproduces the retained scalar metric and its complete optical response. The reference-relative seconds therefore quantify a conditional transfer; they do not by themselves identify a unique underlying state. The finite refit makes this distinction concrete without changing the injected preparation.

BHSM determines the research question and interpretation. The R1 action is an effective representation with its own independent rolling scalar; a unique physical reduction of BHSM cap/interface data to that action requires additional constitutive and embedding data. Those source conditions are derived separately in Supplementary Note S4. The present paper establishes the forward response and its information limits at the level actually specified. Its profiles are controlled examples; no observed lens is fitted and no empirical detection is claimed.

2. Envelope State and Environmental Preparation

The incoming marker probes the event; it does not select the envelope reference state. The conditional state consists of the frozen R1 branch, the finite local density excess and the regular local boundary branch specified below. The earlier source-free target state is retained as a separate null control in Appendix A. Homogeneous geometry, local matter and the local scalar response are accounted for once.

2.1. Incident light, finite body and environment

The lens and source redshifts are $z_l = 0.3$ and $z_s = 1$. The incoming test light is a narrow transient marker with identical emission phase, spectrum, polarization and physical transverse source coordinate in the effective response and its reference. The observed reference wavelength is 550 nm. The fixed source displacement is $s = (0.0014576633674174603, 0)$ Mpc. The light intensity is taken in the test-field limit, so its gravitational backreaction is neglected. The observables are image directions and observer proper-time differences between copies of that marker.

The local body in the conditional envelope is prescribed as a finite excess density above the homogeneous environment. The effective response and fixed-body reference use this same profile:

$$\rho(r) = \frac{15M}{8\pi R^3}\left(1 - \frac{r^2}{R^2}\right) \quad (0 \le r \le R), \qquad M = 10^{11} M_\odot, \quad R = 3 \text{ kpc}. \tag{1}$$

It vanishes outside the body. This is the quadratic-density Tolman–VII profile [11], used as a controlled spherical mass distribution. Define

$$m_g = \frac{GM}{c^2}, \qquad f_M(r) = \frac{M(<r)}{M} = \begin{cases} \frac{1}{2}(5x^3 - 3x^5), & x = r/R \le 1, \\ 1, & r \ge R. \end{cases} \tag{2}$$

For the effective scalar response, the density is prescribed and the leading pressure adjustment is calculated separately. We do not reuse the exact Einstein-fluid pressure as an exact modified-gravity solution.

The physical Jacobi maps are independently initialized for observer–lens, observer–source and lens–source propagation. They share one background, the same transported screen basis, and frequency normalization. At the reference state their scalar values are

$$(D_{ol}, D_{os}, D_{ls}) = (945.162839373,\ 1689.312239687,\ 1076.994348060)\ \text{Mpc}. \tag{3}$$

The main body is excluded from these maps and enters the local encounter once. The local response calculation holds the maps fixed, isolating the encounter contribution. Its fixed-body reference uses the same R1 geometry; it is not a fit of a separate complete ΛCDM cosmology.

3. Action, Normalization, Local Source and Boundary Data

3.1. Conditional BHSM state representation

The question is how an envelope state determines a light transfer, and which state information that transfer identifies. At the effective level used here, the state specifies the rolling values H, v, d, the cubic normalization λ, the finite matter source and the regular boundary branch. The normalized action below makes their source and metric applications explicit. This is the state-conditioned forward problem used throughout the paper.

A unique microscopic map from BHSM cap/interface variables into this independent scalar action is not assumed. The cap momentum and source pullback, including their normalization and complement dependence, are worked out in Supplementary Note S4. They specify additional physical selection data rather than another adjustable lensing contribution. The present light calculation measures properties of the conditional metric response. Its equivalence to a conventional effective mass is then a statement about what those observations can identify.

3.2. Effective action and frozen R1 normalization

We use signature $(-,+,+,+)$ and $M_{\rm Pl}^{-2} = 8\pi G$. In the action and local field equations, $c = \hbar = 1$. The conditional R1 representation uses the retained scalar–tensor action

$$S = \int \sqrt{-g}\,\mathrm{d}^4x \left[\frac{M_{\rm Pl}^2}{2}R - \frac{1}{2}(\nabla T)^2 + \frac{(\nabla T)^2 \Box T}{\lambda} - V_0 + \mu^3 T + \mathcal{L}_m\right], \qquad \lambda = \Lambda^3 > 0. \tag{4}$$

The field T is an independent scalar. It is distinct from the target field η and from a metric conformal coordinate. For example, $\gamma_{ij} = a^2 e^{2\psi_{\rm geo}}\bar{\gamma}_{ij}$ gives $\psi_{\rm geo} = -\Psi$ to linear order in Newtonian gauge; it does not supply another independent scalar in addition to the metric.

The dimensionless background convention is $\widehat{T} = T/M_{\rm Pl}$, $\widehat{v} = \dot{\widehat{T}}/(M_{\rm Pl}H_0)$, $\widehat{d} = \ddot{\widehat{T}}/(M_{\rm Pl}H_0^2)$, $\widehat{\lambda} = \lambda/(M_{\rm Pl}H_0^2)$. The frozen parameters are

$$\begin{aligned}(\Omega_{m0}, \Omega_{r0}, \Omega_{k0}) &= (0.31, 9\times10^{-5}, -0.018),\\ H_0 &= 67.4\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \qquad \widehat{\lambda} = 1, \qquad \frac{\mu^3}{M_{\rm Pl}H_0^2} = 3,\\ \frac{V_0}{M_{\rm Pl}^2H_0^2} &= 2.3986073449708067.\end{aligned} \tag{5}$$

They define one reference branch, not a framework-wide calibration. The stored background solution is reused without refitting. R1 serves as a reproducible fiducial background for this transfer benchmark; no fit to current cosmological observations or claim of phe-

nomenological viability is made here. The reference spatial curvature is $K = -\Omega_{k0}H_0^2/c^2 > 0$. Its unperturbed angular-diameter maps obey $D_{ab} = a_b \sin[\sqrt{K}(\chi_b - \chi_a)]/\sqrt{K}$. The previously computed six-amplitude sensitivity of one normalized axial $n = 2$ harmonic is retained as a separate propagation control; its amplitudes are zero in the reference maps used for the local comparison. Its homogeneous scalar current,

$$J = \dot{\bar{T}} + \frac{6H\dot{\bar{T}}^2}{\lambda}, \qquad \dot{J} + 3HJ = \mu^3, \tag{6}$$

also provides a sign and normalization check on the local variation.

3.3. Reduction from the retained effective action

Varying Equation (4) gives

$$\Box T + \frac{2}{\lambda}\Big[(\nabla_\mu\nabla_\nu T)^2 - (\Box T)^2 + R_{\mu\nu}\nabla^\mu T\nabla^\nu T\Big] - V_T = 0. \tag{7}$$

The Ricci term couples the scalar response to matter through the same Einstein equation. No independent phenomenological matter coupling is introduced. Write $T = \bar{T}(t) + \pi_T(r)$ and freeze $v = \dot{\bar{T}}$, $d = \ddot{\bar{T}}$ and H at the lens epoch. After subtracting the homogeneous equations, retain leading weak metric potentials and quasi-static derivatives, including the nonlinear spatial Hessian responsible for screening. Set

$$b_T = \frac{2v^2}{\lambda}, \qquad C = 1 + \frac{4(d + 2Hv)}{\lambda} - \frac{2v^4}{M_{\rm Pl}^2\lambda^2}. \tag{8}$$

The leading common constraints are

$$2M_{\rm Pl}^2\Delta\Phi = \delta\rho - b_T\Delta\pi_T, \qquad \Phi = \Psi, \tag{9}$$

$$C\Delta\pi_T - \frac{2}{\lambda}\Big[(\Delta\pi_T)^2 - (\partial_i\partial_j\pi_T)^2\Big] = -\frac{b_T}{2M_{\rm Pl}^2}\delta\rho. \tag{10}$$

The last term of C follows from eliminating the metric constraint. The equal potentials are a result at the retained order; they are not an exact assertion about a fully nonlinear space-time.

At $z_l = 0.3$ the frozen solution yields

$$\begin{aligned} H/H_0 &= 1.1836105772, & \widehat{v} &= 0.2377644264, \\ \widehat{d} &= 0.2484715891, & b_T/M_{\rm Pl} &= 0.1130638449, \\ C &= 4.2388585594. \end{aligned} \tag{11}$$

The physical b_T has mass dimension one. In numerical units $\widehat{\lambda} = 1$ is fixed; it is not a new local adjustable length.

3.4. Regular finite-body branch

For spherical symmetry the scalar equation integrates to

$$Cr^2\pi_T' - \frac{4}{\lambda}r(\pi_T')^2 = -b_T GM(<r). \tag{12}$$

The origin is regular and carries no additional scalar charge. The root continuously connected to the decaying weak-source solution is

$$\pi_T' = -\frac{2b_T GM(<r)}{r^2\left[C+\sqrt{C^2+16b_T GM(<r)/(\lambda r^3)}\right]}. \tag{13}$$

It has $\pi_T' \propto r$ at the center and no interface delta-function source at R.

3.5. Executed source and material-boundary application

The source consumed by this reduction can be made explicit. Define an additional metric energy source ρ_Σ and independent scalar Euler source J_T by the pre-elimination rows $2M_{\rm Pl}^2\Delta\Phi = \rho+\rho_\Sigma - b_T\Delta\pi_T$ and $A\Delta\pi_T - 2[(\Delta\pi_T)^2 - |\,\mathrm{Hess}\,\pi_T|^2]/\lambda + b_T\Delta\Phi = J_T$, where $A = C + b_T^2/(2M_{\rm Pl}^2)$. The actual reduced source and spherical flux are

$$J_{\rm eff} = J_T - \frac{b_T}{2M_{\rm Pl}^2}(\rho+\rho_\Sigma),$$
$$Cr^2\pi_T' - \frac{4}{\lambda}r(\pi_T')^2 = \int_0^r s^2 J_{\rm eff}(s)\,\mathrm{d}s. \tag{14}$$

For the declared action there is no extra layer term: $\rho_\Sigma = J_T = 0$, while the volume source remains nonzero. For a proper-radial thin layer with coefficients σ_Σ and $j_{T,\Sigma}$, its retained weak-metric jump would instead be

$$[Cr^2\pi_T' - 4r(\pi_T')^2/\lambda]_{R^-}^{R^+} = R^2\left(j_{T,\Sigma} - \frac{b_T\sigma_\Sigma}{2M_{\rm Pl}^2}\right). \tag{15}$$

These are source applications of the original coupled equations; a surface charge is not an independently adjustable lensing amplitude. An additional anisotropic interface stress would also consume the corresponding metric slip equation.

Using the frozen coefficients, the computed surface section is $\pi_T'(R)/(M_{\rm Pl}H_0) = -2.119825778845613\times10^{-4}$. In $M_{\rm Pl}=H_0=c=1$ units, with radius in c/H_0, an independent density integration gives $\int_0^R r^2 J_{\rm eff}\,\mathrm{d}r = -1.216417408185084\times10^{-13}$. The maximum relative nonlinear-flux residual at $r/R = 0.1, 0.5, 1, 2, 10$ is 4.15×10^{-16}; the original integrated metric Gauss residual is zero at printed precision. Independent interior quadratic and exterior rational roots give a relative material-boundary flux mismatch 6.23×10^{-16}. The separate exact-GR material control has continuous same-oriented extrinsic curvature and no material layer. That material worldtube is distinct from the cosmological cap seam, and its zero layer does not remove the nonzero volume source. The receipt is `artifacts/strong_lensing_bhsm_integration_20261020/source_application.json`. Supplementary Note S4 derives the cap/source pullback and shows constructively why the leading normalization does not select a unique microscopic completion.

4. Metric and Optical Encounter Response

The common metric constraint gives

$$\Phi' = \Psi' = \frac{GM(<r)}{r^2} - \frac{b_T}{2M_{\rm Pl}^2}\pi_T', \qquad F(r) \equiv \frac{\delta\Phi'}{\Phi_N'} = \frac{2\epsilon_\infty}{1+\sqrt{1+r_V^3 f_M(r)/r^3}}, \tag{16}$$

where

$$\epsilon_\infty = \frac{b_T^2}{2M_{\rm Pl}^2 C}, \qquad r_V^3 = \frac{16 b_T GM}{\lambda C^2}. \tag{17}$$

The last expression uses $c = 1$. The numerical conversion is

$$r_V^3 = 16(b_T/M_{\rm Pl})m_g(c/H_0)^2/(\widehat{\lambda}C^2).$$

Deep in the exterior screened region $F(r)$ approaches $2\epsilon_\infty(r/r_V)^{3/2}$; at large radius it approaches ϵ_∞. The potential itself still decays. This is the usual cubic screening mechanism [4,5], evaluated here with the frozen normalization and a regular extended body.

The corresponding leading hydrostatic correction is

$$\delta p(r) = \int_r^R \rho(s)\,\delta\Phi'(s)\,\mathrm{d}s, \qquad \delta p(R) = 0. \tag{18}$$

It is positive and changes the central Newtonian pressure by 3.56959×10^{-6} fractionally. This supports the specified leading-order matter configuration; it does not construct an exact modified Tolman–Oppenheimer–Volkoff sequence or prove radial stability. Holding density fixed and changing its required pressure does not impose an independently fixed common material equation of state on both models.

4.1. Projection into the optical encounter

Restore c and interpret Φ as the dimensionless metric potential. For physical impact radius $b_{\rm imp}$ the additional deflection is

$$\delta\hat{\alpha}(b_{\rm imp}) = 4b_{\rm imp}\int_0^\infty \frac{\delta\Phi'(\sqrt{b_{\rm imp}^2+z^2})}{\sqrt{b_{\rm imp}^2+z^2}}\mathrm{d}z, \tag{19}$$

$$\delta\hat{\psi}(b_{\rm imp}) = \int_R^{b_{\rm imp}} \delta\hat{\alpha}(s)\,\mathrm{d}s. \tag{20}$$

The arbitrary zero of potential does not affect relative delays. We calculate the gradient and Hessian consistently from this same projected potential. A local force ratio is not substituted for a projected photon deflection ratio.

The leading stationary delay change between image i and reference image 0 is

$$\delta\Delta t_{i0} = -\frac{1+z_l}{c}\left[\delta\hat{\psi}(b_i) - \delta\hat{\psi}(b_0)\right]. \tag{21}$$

The finite result also resolves the image positions in Equation (23). The projection freezes the local coefficients during the encounter and extends the isolated local solution through the thin plane. Matching to the linear perturbation is assumed in an overlap region $r_V \ll r \ll c/H_l$; a global closed-cosmology solution has not been constructed.

4.2. Which state coefficients enter the transfer?

The state dependence is explicit in $b_T = 2v^2/\lambda$ and $C = 1 + 4(d+2Hv)/\lambda - b_T^2/(2M_{\rm Pl}^2)$. At fixed body, λ and physical radius, define $S_r = \sqrt{1 + r_V^3 f_M(r)/r^3}$. Differentiating the original force gives

$$\frac{\partial \ln\delta\Phi'}{\partial \ln b_T} = \frac{3}{2} + \frac{1}{2S_r}, \qquad \frac{\partial \ln\delta\Phi'}{\partial \ln C} = -\frac{1}{S_r}. \tag{22}$$

Deep screening suppresses the local force sensitivity to C. Projection in Equation (19) takes positive weighted averages of these coefficients. For the reference image impacts, ordered positive, central and negative, the projected logarithmic derivatives are

$$\partial_{\ln b_T} \ln \delta\hat{\alpha} = (1.525461, 1.516634, 1.522208),$$
$$\partial_{\ln C} \ln \delta\hat{\alpha} = (-0.050923, -0.033268, -0.044417).$$

The outer field therefore retains nonzero C sensitivity even though the impact lies well inside r_V.

These derivatives concern the extra deflection; derivatives divided by the total deflection carry its additional small fractional contribution. They quantify the conditional state-to-transfer relation without changing the frozen prediction. A particular physical state variation must also satisfy the action's background and boundary constraints and be composed with its H, v, d to b_T, C map. Independently varying the two reduced coefficients is not asserted to produce another globally matched R1 state. Section 6 addresses the separate question of identifying those coefficients from light when the ordinary mass distribution is allowed to change.

5. Image Directions and Observer Delays

5.1. Image directions and observer arrival-time differences

Let $x = D_{ol}\theta$, $\beta = D_{os}^{-1}s$ and $\hat{\alpha} = \nabla_x \hat{\psi}$, where $\hat{\psi}$ has length units. The common thin-lens equations are

$$\beta = \theta - D_{os}^{-1} D_{ls} \hat{\alpha}(D_{ol}\theta), \tag{23}$$
$$ct(\theta) = \frac{1}{2}(\theta - \beta)^T \mathsf{T}(\theta - \beta) - (1 + z_l)\hat{\psi}(D_{ol}\theta) + f(\beta),$$
$$\mathsf{T} = (1 + z_l) D_{ol}^T D_{ls}^{-1} D_{os}. \tag{24}$$

The maps must satisfy reciprocity so that T is symmetric [2]. A common $f(\beta)$ cancels in relative delays. Image positions are resolved again after changing the encounter.

For smooth geometric variations, fixing s implies $\delta\beta = -D_{os}^{-1}\delta D_{os}\beta$; fixed physical and fixed angular source prescriptions cannot be imposed simultaneously. At fixed measured redshift, the stationary first variation is

$$c\delta t = \frac{1}{2} q^T \delta\mathsf{T} q - q^T \mathsf{T}\delta\beta - (1 + z_l)\left[\hat{\alpha}^T \delta D_{ol}\theta + \delta\hat{\psi}|_x\right] + \delta f, \qquad q = \theta - \beta. \tag{25}$$

The image-position variation cancels at a stationary image. In the local R1 comparison $\delta D_{ab} = \delta\beta = 0$ and the nonzero intrinsic term is retained. Appendix B records the measured-redshift convention for comparisons that vary the maps.

5.2. Evaluated response and reference-relative offsets

The screened branch predicts $r_V = 212.0292$ kpc and $\epsilon_\infty = 0.001507886$, or a 0.1507886% asymptotic force enhancement. At $r = R = 3$ kpc the local force enhancement is 5.06706 parts per million (ppm), whereas the deflection enhancement at the same impact radius is 11.7467 ppm. The ray samples the outer, less screened field as well as the field near closest approach.

Figure 1 compares the radial response and the two delay changes. The nonzero response follows from the rolling scalar already in Equation (4). It does not alter the earlier constant-target null result in Appendix A. An independent singular isothermal control, with velocity dispersion 220 km s^{-1} and the same reference source, retains images at $+1.067883$ and -0.711922 arcsec and a 17.094532-day delay. That analytic control validates the optical

Table 1. Leading-order image positions for the effective response and its fixed-body reference. Image 0 is the positive outer image, image 1 the central image, and image 2 the negative outer image. Printed digits specify reproducible model values, not absolute astrometric accuracy.

Image	Reference (arcsec)	R1 response (arcsec)	Offset (μas)
0	+0.835476034	+0.835482157	+6.1233
1	−0.082161940	−0.082161152	+0.7881
2	−0.657495525	−0.657499874	−4.3486

Table 2. Relative delays for the same source marker, with image 0 as reference. The offsets are relative to the specified fixed-body reference, with the physical light preparation held fixed.

Observable	Fixed-body reference	R1 response	Offset
Δt_{10} (days)	21.716076566	21.716377020	25.9593 s
Δt_{20} (days)	14.271452387	14.271552491	8.6490 s
$\Delta t_{10}/\Delta t_{20}$	1.521644467	1.521654847	6.8212 ppm

conventions; the finite body in Tables 1 and 2 is the local boundary problem used for the evaluated response.

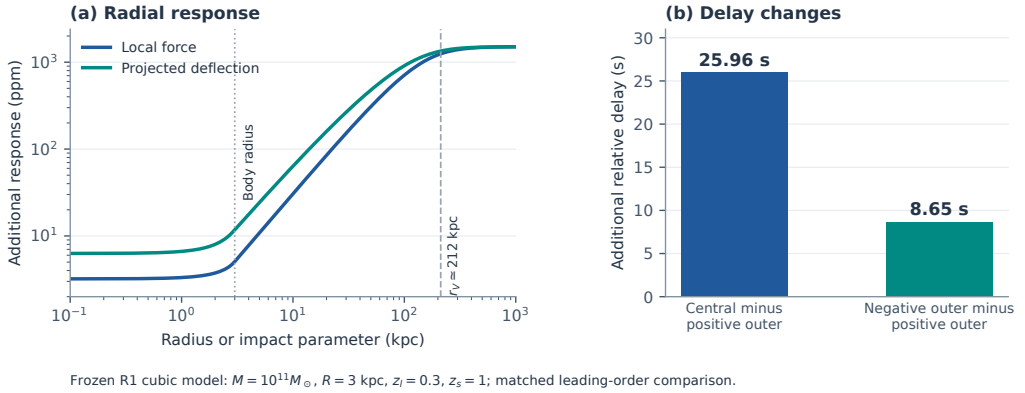


Figure 1. R1 envelope response relative to the fixed-body reference. Left: additional local force and projected deflection as fractions of their fixed-body reference values. The distinction reflects integration along the ray. Right: reference-relative offsets in the two delays for the fixed source preparation. Neither panel is an observational detection forecast.

At fixed epoch, $r_V \propto M^{1/3}$. The same frozen background also specifies the redshift dependence, without refitting:

Lens redshift	ϵ_∞	r_V (kpc), $M = 10^{11} M_\odot$
0.0	0.00351132	248.300
0.3	0.00150789	212.029
0.5	0.00086052	191.230
1.0	0.00022832	150.778
1.5	0.00007085	123.011

This is a prediction for the coefficients and force law, not a delay table for identical angular sources at these different redshifts.

The supplied reference-run outputs give the following implementation checks. The integrated scalar flux has maximum relative residual 7.9×10^{-16}. A direct line-of-sight quadrature and an independent angular-substitution quadrature agree on the additional deflection to 4.5×10^{-13} relative. The image equation residual is below 8.5×10^{-22} rad.

Differentiating the projected potential and deflection independently gives relative discrepancies below 2.0×10^{-9} and 2.4×10^{-12}, respectively. Finite re-solved and stationary first-order delay changes differ by 0.00016 s and 0.000056 s. These are numerical and perturbative checks within the model, not bounds on omitted physics. In the current Windows replay, the corresponding flux residual is 8.30×10^{-16} and the deflection-derivative discrepancy is 3.19×10^{-12}; the remaining differences are recorded with their numerical tolerances in the replay receipt. The supplied and replay outputs remain separate records.

6. What the Light Transfer Identifies

6.1. Positive effective-mass representation of the same response

The retained metric and optical response admit an explicit conventional representation. Define an effective additional enclosed mass by

$$\delta M_{\rm eff}(r) = M f_M(r) F(r), \qquad \delta\rho_{\rm eff}(r) = \frac{M}{4\pi r^2}\frac{\mathrm{d}}{\mathrm{d}r}[f_M(r)F(r)]. \tag{26}$$

This density is nonnegative. To see this, put $S = \sqrt{1 + r_V^3 f_M / r^3}$ and $n_M = \mathrm{d}\ln f_M / \mathrm{d}\ln r \geq 0$. Then

$$\frac{\mathrm{d}\ln(f_M F)}{\mathrm{d}\ln r} = \frac{1}{2}\Big[(1+S^{-1})n_M + 3(1-S^{-1})\Big] \geq 0. \tag{27}$$

The ordinary Poisson field of $M(<r) + \delta M_{\rm eff}(r)$ has exactly the same $\Phi = \Psi$ as Equation (16) within the retained approximation. It therefore reproduces the full deflection field, relative potential, images and delays, including multiple source planes in the same geometry. Its asymptotic extra mass is

$$\delta M_{\rm eff}(\infty) = \epsilon_\infty M = 1.50788624 \times 10^8 M_\odot. \tag{28}$$

The rank-one multiple-source residual discussed below distinguishes the scalar from the restricted Tolman–VII parameter family, not from all conventional matter distributions. Because slow matter and light respond to the same retained potentials, ordinary dynamical data do not automatically break this equivalence either. Independent restrictions on the matter/environment model or an observable outside this retained equivalence would be needed to distinguish those representations. No such independent likelihood is asserted here, and such a distinction is not required for the envelope-transfer calculation.

The effective mass in Equation (26) represents the same retained contribution as the scalar correction in Equation (16). One uses either the explicit scalar correction or its effective-mass representation in the encounter, not their sum. The construction proves equivalence of the retained metric and optical response. The physical pullback of microscopic BHSM surface variables into the normalized effective source is specified separately in Supplementary Note S4.

6.2. Single-source constraints and nuisance freedoms

For an observable residual $\delta y = y - y_0$, write

$$\delta y = B\,\delta a + N\,\delta\nu + \epsilon, \qquad W\mathsf{C}W^T = I, \qquad P = I - (WN)(WN)^+. \tag{29}$$

Here B is a physical response, N the stated nuisance response, C a specified covariance, and $+$ a Moore–Penrose inverse. The rank of PWB concerns distinguishability in that nuisance model. A zero direction is unconstrained by the corresponding likelihood; a Fisher pseudoinverse does not turn it into a zero error bar. We take C positive definite. For the screened comparison, the single column is a fixed reference-relative response

template; rank one denotes a surviving residual direction, not identification of a freely varied fundamental scalar parameter.

Using source position, body mass and radius, and a compensated mass-sheet scale as nuisances, the single R1-response column has projected rank one if the body center is fixed. Adding both center coordinates gives rank zero. The raw delay shifts in Table 2 are therefore not nuisance-independent observables for this configuration. The rank calculation uses 0.001 arcsec and 0.1 day as numerical whitening scales, not as demonstrated observational precisions.

The ordinary mass-sheet transformation,

$$\psi_\lambda(\theta) = \lambda\psi(\theta) + \frac{1-\lambda}{2}|\theta|^2, \qquad \beta_\lambda = \lambda\beta, \qquad \Delta t_\lambda = \lambda\Delta t, \tag{30}$$

cancels from the ratio of two delays. It does not remove source, centering, radial-profile, or independent line-of-sight freedoms [8]. Consequently the 6.8212 ppm ratio change is a fixed-preparation prediction, not a unique diagnostic of a distinct representation.

For the separately computed axial smooth-map response, the images of a softened isothermal model depend only on $\beta = s/D_{os}$, $\theta_E = b_{\rm phys}D_{ls}/D_{os}$, and $\theta_c = r_c/D_{ol}$, while all delays share the factor $(1+z_l)D_{ol}D_{os}/(cD_{ls})$. Varying the three scalar maps can therefore be reproduced by these source/profile changes and a delay scale. The saved six-anchor response accordingly has zero projected rank in the unrestricted nuisance model. This analytic factorization concerns that family; it is not a universal theorem that cosmological lensing information always vanishes.

6.3. A finite nonlinear conventional refit

A local rank calculation can be tested by a finite inverse application. We take the saved R1 images and delays as a deterministic injection and fit an ordinary Tolman–VII body with mass, radius, source position, lens center and one physical convergence sheet free. The injected marker, state, geometry and source remain the original preparation; fitted source coordinates are inverse-model nuisances. The fitted encounter contains the sheet once, while the propagation maps remain fixed. No independent delay scale is introduced.

For each of the three observed axial image positions, the original rows are $\theta_i - \beta - q\hat{\alpha}_{M,R}[D_{ol}(\theta_i - \theta_c)] - \kappa(\theta_i - \theta_c) = 0$, with $q = D_{ls}/D_{os}$. The two relative Fermat arrival rows use

$$\hat{\psi}(x) = \hat{\psi}_{M,R}(x - D_{ol}\theta_c) + \frac{\kappa}{2D_{ol}q}|x - D_{ol}\theta_c|^2. \tag{31}$$

All three image branches are re-solved after fitting. This is a bounded five-coordinate fit to five axial observables, a deliberately saturated local example whose Jacobian is nonsingular. It demonstrates finite absorption in this example; it is not a global uniqueness or detection claim. The free-center original lens and stationarity residuals are 4.24×10^{-22} rad and 1.30×10^{-18} Mpc. Allowing the ordinary model to adjust therefore removes the original 25.96 s and 8.65 s fixed-body offsets to numerical precision in this preparation. Fixing the center leaves a small finite residual; its significance would require an actual observing likelihood and independently justified lens priors. Supplementary Note S7 gives the original equations, parameter bounds, mass-sheet symmetry and independent-root checks.

6.4. A bounded multiple-source test

A supplemental design uses one body and three sources at $z_s = (0.8, 1.2, 1.8)$, with source offsets $0.1\theta_{E,s}$ at angles $(0.15, 1.1, 2.2)$ rad. These choices were fixed before applying the screened response. The nuisance model includes independent source positions and delay scales, a shared mass, radius and center, and shared physical convergence and two

Table 3. Finite conventional fits to the saved R1 injection. Entries are numerical mismatches after re-solving image branches, not observational errors or sensitivity forecasts.

Quantity	Center fixed	Center free
$\delta \ln M$	3.9960×10^{-6}	4.0206×10^{-6}
$\delta \ln R$	3.9827×10^{-7}	4.1234×10^{-7}
$\delta\beta$ (arcsec)	4.3495×10^{-9}	6.8148×10^{-9}
θ_c (arcsec)	0	1.7743×10^{-9}
κ	9.9469×10^{-6}	9.9229×10^{-6}
Maximum image mismatch (μas)	1.0143×10^{-4}	8.74×10^{-11}
First delay mismatch (s)	$+0.0033064$	$+9.21 \times 10^{-10}$
Second delay mismatch (s)	-0.0050312	-1.53×10^{-10}

shear components. The resulting scalar column retains rank one; adding independent source-plane convergence columns leaves rank one as well.

This surviving component is small: its norm is 0.1202% of the raw whitened response in the expanded nuisance model. In the same illustrative weighting, the residual coordinates reach approximately 6 nanoarcsec and the residual delays 0.086 s. The explicit 24 observable weights and response arrays are supplied with the code. These tidal nuisance fields are phenomenological controls. This test does not solve the nonlinear scalar equation for a nonspherical galaxy and its environment, and it is not a signal-to-noise forecast. Its rank is specific to the finite radial profile family.

7. Robustness of the Transfer Calculation

7.1. A smooth profile with the same mass and half-mass radius

The body is a controlled input, so the response must be tested against its profile choice. We compare the compact Tolman–VII body with the standard smooth Plummer sphere [12], matching total three-dimensional mass and three-dimensional half-mass radius, rather than fitting a desired deflection. The reference half-mass radius is $r_h = 1.92941634649$ kpc. The Plummer scale is fixed by $a = r_h\sqrt{2^{2/3} - 1} = 1.47874508325$ kpc, with

$$\rho_P(r) = \frac{3Ma^2}{4\pi(r^2 + a^2)^{5/2}},$$
$$M_P(< r) = \frac{Mr^3}{(r^2 + a^2)^{3/2}}, \qquad \hat{\alpha}_P(b) = \frac{4m_g b}{b^2 + a^2}. \tag{32}$$

This profile has a regular center and finite total mass, with no hard material edge. Its consistent potential is $\hat{\psi}_P = 2m_g \ln[(b^2 + a^2)/(2a^2)]$. The same original scalar flux, regular root and metric constraint are applied to its enclosed mass. The background, clock and maps remain fixed. Both profiles have the same reference r_V because their total mass and R1 state are equal.

Eight paired applications use the two profiles and four settings: reference, total mass increased by 20%, half-mass radius increased by 20%, or physical source displacement increased by 20%. Only one ordinary input changes at a time, and the change is shared by the response and its same-profile reference. These are sensitivity controls, with no assigned probability or empirical uncertainty prior. All eight applications retain three images.

At matched M, r_h, changing the ordinary profile changes the two reference delays by approximately -167125 s and -124500 s, while changing the R1 offsets by -3.5083 s and $+0.9710$ s. The former are deliberate model changes, about 6438 times the largest reference R1 offset; they are not an estimated error on an observed lens. This shows why mass and

Table 4. Re-solved reference-relative delay offsets for two declared profiles. Each pair uses common preparation, state and ordinary inputs. The finite input changes are controls, not fitted uncertainties.

Shared control	Tolman–VII offsets (s)	Plummer offsets (s)
Reference	25.9593, 8.6490	22.4510, 9.6200
M increased 20%	32.6999, 9.8907	28.6743, 10.7843
r_h increased 20%	24.2697, 9.9458	20.2722, 10.1974
Source offset increased 20%	26.8452, 10.3945	23.5110, 11.5634

one size alone do not fix the transfer. A ray samples the entire field, even though both profiles share the same scalar force at their half-mass radius.

The positive effective-mass construction applies to both profiles. Independent direct line-of-sight projection and re-solved images agree with their scalar representations within 5.56×10^{-17} arcsec and 6.14×10^{-10} s. The maximum original scalar-flux residual across the eight applications is 7.25×10^{-16} and the maximum image-equation residual is 2.55×10^{-21} rad. Both baseline profiles retain projected rank one with a fixed center and zero with a free center in the declared linearized nuisance family. Thus profile smoothness changes the quantitative response while preserving its retained conventional representation. Supplementary Note S6 gives the profile normalization, support and independent checks.

7.2. The ordinary finite-body control

For the conventional control, a regular Einstein-fluid interior can be matched to Schwarzschild. Its compactness is $m_g/R = 1.59514 \times 10^{-6}$. The leading optical application above uses the weak-field projection of this density.

7.3. A common second-order ordinary-gravity control

To test the effect of the first omitted conventional deflection, we expand the exact static GR optical scattering integral of the same Tolman–VII body through second post-Minkowskian order (2PM). Put $\varepsilon_g = m_g/R$, $y = b_{\rm imp}/R$ and $t = \sqrt{1-y^2}$. The additional radial deflection is

$$\delta\hat{\alpha}_{\rm GR2}(b_{\rm imp}) = 2\varepsilon_g^2 I(y), \tag{33}$$

$$I(y) = \begin{cases} \dfrac{15(\arcsin y - yt)}{4y^2} + y\left(\dfrac{15}{2}t - 4t^5 - \dfrac{48}{7}t^7\right), & 0 < y < 1, \\ \dfrac{15\pi}{8y^2}, & y \geq 1. \end{cases} \tag{34}$$

At the center $I(0) = 0$ and $I'(0) = -6/7$; the interior coefficient includes the leading fluid pressure and the conversion from areal radius to invariant asymptotic impact. At and outside the surface it matches the standard $15\pi m_g^2/(4b_{\rm imp}^2)$ exterior deflection [13]. The consistent plane potential is

$$\delta\hat{\psi}_{\rm GR2}(b_{\rm imp}) = 2\varepsilon_g^2 R J(y), \qquad J'(y) = I(y), \qquad J(1) = 0. \tag{35}$$

The supplied implementation evaluates I, I' and J analytically.

The R1 response and fixed-body reference receive the identical GR2 potential, gradient and Hessian, and both image systems are re-solved. The R1 coefficients and leading scalar profile are held fixed. The scattering impact is identified with the physical thin-plane impact in this control. This checks robustness to the common GR term; it does not compute the missing scalar–GR higher-order terms or a complete finite-distance 2PM cosmological arrival functional. The replay command is

```
python scripts/run_strong_lensing_gr2_control.py
```

7.4. Result of the common GR correction

At the original two exterior-image impacts, the GR2 term is 3.68 and 4.68 ppm of the leading conventional deflection. Its inclusion changes the conventional image coordinates by $(+1.35456, -0.01519, -1.72123)$ microarcsec and the conventional relative delays by $(+8.28473, +2.55245)$ s. The absolute correction is therefore material at the scale of the calculated reference-relative image shifts.

The response-minus-fixed-body-reference offsets are considerably more stable under this common-term check.

Table 5. Robustness of the reference-relative R1 offsets under the same GR2 correction in both calculations. The final column is the change in the reference-relative offset, not the absolute GR correction.

Response offset	Leading calculation	Common GR2 added	Change
$\delta\Delta t_{10}$ (s)	25.95925815	25.95932714	$+68.99$ μs
$\delta\Delta t_{20}$ (s)	8.64901938	8.64900924	-10.14 μs
$\delta\theta_0$ (μas)	$+6.12327625$	$+6.12326934$	-6.91×10^{-6} μas
$\delta\theta_1$ (μas)	$+0.78810716$	$+0.78810711$	-5.58×10^{-8} μas
$\delta\theta_2$ (μas)	-4.34859153	-4.34858643	$+5.09 \times 10^{-6}$ μas

The two relative changes in the reference-relative delay offsets are $+2.66$ and -1.17 ppm. The delay-ratio response remains 6.8212 ppm. Thus the nonzero reference-relative response survives this specific common-term control without changing its quoted precision in seconds or microarcseconds.

An independent mixed stationary-action derivative reproduces the change in the reference-relative delay offsets within 0.7 ns; this checks the small drift without subtracting four re-solved delays.

The analytic correction was also compared with direct null-geodesic scattering in the solved static GR fluid and exterior metric. At the three reference impacts, the exact deflection minus the 1PM and 2PM terms has magnitude below 1.8×10^{-16} rad for this body's compactness $\varepsilon_g = 1.59514 \times 10^{-6}$. A compactness extrapolation independently recovers the interior 2PM coefficient, and direct integration verifies $J' = I$. The latter checks establish the implemented ordinary-GR scattering coefficient, not a physical error bound for the scalar model or its cosmological embedding. The full corrected coordinates, delays and checks are supplied in `gr2_control.json` and the review outputs.

7.5. A distributed encounter with the same physical endpoints

The thin plane can be checked by distributing the same retained field along the ray. Use lens-scaled conformal coordinates $z = a_l(\chi - \chi_l)$, $x = a_l x_{\rm com}$ and $\kappa_K = K/a_l^2$, where $a_l = (1 + z_l)^{-1}$. The observer and source lengths are $\ell_o = a_l\chi_l$ and $\ell_s = a_l(\chi_s - \chi_l)$; the fixed source endpoint is $X_s = (a_l/a_s)s$. These are determined by the original closed reference and its physical distance maps. Define $S_\kappa(d) = \sin(\sqrt{\kappa_K}d)/\sqrt{\kappa_K}$, $C_\kappa(d) = \cos(\sqrt{\kappa_K}d)$ and the background generating function

$$G_d(x_1, x_2) = \frac{C_\kappa(d)(x_1^2 + x_2^2) - 2x_1x_2}{2S_\kappa(d)}. \tag{36}$$

For a common half-window L, the retained paraxial arrival action is

$$ct[x] = \frac{1}{a_l}\Bigg\{ G_{\ell_o - L}[0, x(-L)] + G_{\ell_s - L}[x(L), X_s] + \int_{-L}^{L}\left[\frac{x'^2 - \kappa_K x^2}{2} - 2\Phi(\sqrt{x^2+z^2})\right]\mathrm{d}z\Bigg\}. \tag{37}$$

Its original ray equation and endpoint rows are

$$\begin{aligned} x'' + \kappa_K x &= -2\partial_x \Phi, \\ x'(-L) &= \frac{C_\kappa(\ell_o - L)}{S_\kappa(\ell_o - L)} x(-L), \\ x'(L) &= -\frac{C_\kappa(\ell_s - L)}{S_\kappa(\ell_s - L)} x(L) + \frac{X_s}{S_\kappa(\ell_s - L)}. \end{aligned} \tag{38}$$

The individual exterior wavefront curvatures are fixed by those physical endpoint data. At the plane their sum is $D_{os}/(D_{ol}D_{ls})$ and the source term is s/D_{ls}, so they recover the original thin-plane preparation. They are not a freely chosen split of that sum. The factor $1/a_l = 1 + z_l$ retains the same observer clock.

Both the fixed-body and R1 fields are solved on each window, using all three noncritical ray branches. Their distributed actions are compared with the thin-plane/Born projection of the same windowed fields. At $L = 20$ Mpc, the same-window thin-plane offsets are

Table 6. Distributed-path change of the reference-relative delay offsets compared with the same-window thin plane, evaluated by direct arrival subtraction. This is a control of the retained frozen weak-field optical action.

Half-window L (Mpc)	Change of $\delta\Delta t_{10}$ (ms)	Change of $\delta\Delta t_{20}$ (μs)
0.21203	−0.46019	+1.5548
0.84812	−0.57857	+1.1474
4.24058	−0.61238	+1.0321
20.00000	−0.61905	+1.0056

25.95922277 s and 8.64900576 s; the distributed offsets are 25.95860372 s and 8.64900676 s. Tighter collocation and potential quadrature change the latter offsets by less than 0.3 ns. These are refinement figures for that subtraction. A separate on-shell derivative integrated along a numerical interpolation between the two fixed optical fields gives the preferred finite-response changes −0.61904348 ms and +1.00640532 microseconds. They agree with the raw subtraction within 7 ns; the interpolation is a numerical integration device, not a physical state variation. These are numerical controls within this defined optical problem.

An analytic exterior-tail bound follows from $0 \le \delta\Phi' \le \epsilon_\infty m_g/r^2$. For two fixed Born impacts b_i, b_0, it gives

$$|\delta\Delta t_{i0}^{\mathrm{tail}}| \le \frac{1+z_l}{c}\,\frac{m_g \epsilon_\infty |b_i^2 - b_0^2|}{L^2}. \tag{39}$$

The sharper integrated bounds at 20 Mpc are 35.04 and 13.47 microseconds for the two extra-field fixed-impact delays. They do not bound shifts caused by re-solving all images, an arbitrary environment, or a changed global state. The new control isolates finite thickness and path response with fixed background, source and clock. It does not add the scalar-dependent field-equation corrections needed for a full higher-order metric prediction. The complete boundary construction, force/action residuals, tail derivation and convergence data are in Supplementary Note S8.

8. Computational Methods and Reproducibility

The calculation uses the fixed manuscript source revision `49a1dab` and records full input hashes. The main reproduction command from the repository root is

```
python scripts/run_strong_lensing_r1_screened.py
```

The coefficient, independent projection, and multiple-source checks are under `scripts/r1_screened_review/`. Numerical results and their units are under `artifacts/strong_lensing_r1_screened_20261020/`. The research package additions and exact commands are integrated with the pinned repository background and optical dependencies. The recorded package-era replay of three portable review scripts reproduced all 305 saved numerical values in that environment. The current integration replayed nine numerical commands, compared 1161 saved numerical values, and passed 16 focused checks without changing the original numerical artifacts. These are separate execution records, not independent physical measurements.

The additional inverse and profile applications are reproduced with `scripts/run_strong_lensing_refit.py` and `scripts/run_strong_lensing_robustness.py`; the distributed optical control uses `scripts/run_strong_lensing_encounter_accuracy.py`. Their outputs are under `artifacts/strong_lensing_acceptance_20261020/`. Four refit, seven profile/coefficient and seven encounter tests check the original applications and independent projections. The distributed-control receipt records its boundary identities, original variational residuals and convergence checks. The complete commands and software versions are in the acceptance checkpoint. The original 305-value portable package check and the current 1161-value replay remain separate records.

The reproducibility manifest identifies versions and hashes rather than equating successful execution with physical validation.

The independent quadrature uses the same physical radial equation but a separately implemented integration variable and root calculation. Its agreement is a reproducibility check, not independent empirical evidence or external human peer review.

9. Discussion: State Inference and Physical Scope

The calculation answers one forward and inverse modeling question. A specified R1 state and finite source determine a regular encounter and its light observables. The coefficient derivatives quantify that state dependence. A positive effective mass represents the same retained metric, and a finite conventional nuisance fit reproduces the injected three-image system. Accordingly, the reference-relative seconds are a calculated response at fixed preparation; assigning them to a unique envelope state requires restrictions on the ordinary lens model and the state-to-source relation. A shared multiple-source profile can retain information that a one-source fit absorbs, but the complete positive effective-mass representation remains available at this order. This is a constructive equivalence within the BHSM transfer program.

The smooth-profile comparison gives the practical scale of this issue. The imposed profile changes alter ordinary delays much more than the state-conditioned offsets. They do not define an empirical prior; they show why a model-conditioned transfer is insufficient on its own for state identification. The finite refit is likewise a saturated local example rather than a forecast. Applying the benchmark to an observed system would require its imaging, time-delay likelihood and independently justified source, mass and environment information. Neither prescribed spherical profile is presented as a measured galaxy, and central-image detectability is not assumed.

An observational extension can use existing light data to establish a baseline for each individual lens. Archival resolved imaging and monitoring light curves would constrain the conventional potential, source variability and time-delay likelihood jointly with the

permitted conditional BHSM response. COSMOGRAIL's long monitoring record provides an example of existing light curves and subsequent observing seasons with which to organize this test [14]. The baseline must retain its mass, source, line-of-sight and calibration uncertainties; a published cosmological posterior already conditioned on a lens model is not a substitute for those light-data likelihoods.

The model family, calibration data and priors would then be fixed before predicting withheld image information or subsequent monitoring data. New observations would test the predictive distribution before entering an updated fit. This procedure could build a progressively constrained optical-response map for each lens, including effective geometry or an interface response when its source law is specified. It does not make the underlying microscopic state unique: the conventional equivalences demonstrated here must remain in the inference. An observed asymmetric lens also requires an appropriate spatial model and matching propagation maps, rather than an imposed spherical three-image fixture. No observed lens has been fitted in the present synthetic benchmark.

The approximation controls have separate meanings. The common GR2 term moves absolute image coordinates by up to 1.72 microarcsec, while its change of the reference-relative image offsets is below 7×10^{-6} microarcsec and its delay-offset change is below 0.1 ms. The distributed-encounter control tests the optical plane within the retained leading action. Its finite-window and path effects are distinct from scalar-dependent field-equation higher orders. The local parameter $H_l r_V/c = 5.64 \times 10^{-5}$ and the 1.1–1.7 $\times 10^{-6}$ fraction of extra deflection beyond 20 Mpc in the formal isolated projection characterize scale separation and tails. They are not complete physical error bounds. A time-dependent material/environmental continuation consumes the specific equation of state or support and constrained outer data derived in Supplementary Note S5.

The leading scalar principal coefficients are positive on the retained branch (Appendix C); its maximum radial characteristic speed squared is approximately 1.474 in metric light-speed units. Photons in this benchmark follow metric null rays. This leading coefficient check establishes neither full coupled stability nor a causal ultraviolet completion. The independent rolling scalar T, target phase and metric coordinate retain their stated normalizations; the earlier source-free target null is preserved in Appendix A.

The BHSM microscopic source question has a precise boundary in this paper. Supplementary Note S4 derives the cap/matcher and Schur source pullback and gives positive same-domain complements with identical leading normalization but different higher-derivative coefficients. A physical cap/interface embedding into canonical T, its cubic vertex, source pairing and complement length/spectrum must therefore be supplied by the framework before a unique microscopic state is assigned to this transfer. This result supplies the defining conditions and their nonuniqueness; it does not select an extra optical amplitude. The present calculation is a reproducible effective benchmark against which that physical source construction can be tested.

10. Conclusions

For the prescribed $10^{11} M_\odot$, 3 kpc body at $z_l = 0.3$ and common source at $z_s = 1$, the fixed R1 action/state produces a regular source, screening radius approximately 212 kpc, reference-relative delay offsets 25.96 s and 8.65 s, image changes approximately 0.79–6.12 microarcsec and a 6.82 ppm delay-ratio change. The state dependence enters explicitly through the normalized scalar mixing and stiffness. These results retain one physical incident-light, clock and propagation prescription throughout.

The same retained metric and light transfer admit a positive conventional effective mass. A finite five-parameter ordinary refit also reproduces the three injected images and two delays to numerical precision; fixing its center leaves millisecond-scale delay

mismatches in the declared profile family. The interpretation of the response therefore depends on the admitted lens freedoms. A smooth Plummer control matched by total mass and half-mass radius gives offsets 22.45 s and 9.62 s while preserving the effective-mass equivalence. The eight paired profile/input controls, common ordinary-GR correction and distributed optical encounter make the benchmark's model and approximation dependencies explicit. The common GR2 changes of the original offsets remain $+69$ and -10 microseconds.

Prepared light constrains an envelope encounter transfer. Identifying that transfer with a particular underlying state consumes the specified action/source map and independently justified mass/environment data. The calculation provides a coherent forward and inverse test problem for that inference, with its physical source conditions and technical continuation derived separately.

Supplementary Materials: The repository supplies Notes S1–S8: local action/source reduction; finite-body GR2; nuisance optics; cap/interface source pullback; exact field/environmental continuation; profile robustness; finite nonlinear refit; and distributed-encounter accuracy control. Their source paths, executable commands, input hashes, numerical outputs and focused checks are mapped in the acceptance checkpoint. The long cap and continuation TeX derivations are retained as technical supplementary source; the historical target and transport calculations remain available.

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Appendix A. The Constant-Target Null Control

The earlier source-free target sector has

$$S_\eta = -\int N\sqrt{h}\, w_0 F(X)\, \mathrm{d}t\, \mathrm{d}^3x, \qquad F(X) = \frac{\kappa_\eta}{2}X + \frac{1}{8}X^4, \qquad X = g^{\mu\nu}\langle D_\mu\eta, D_\nu\eta\rangle, \quad \text{(A1)}$$

with $\eta = (s,z) \in S^6 \subset \mathbb{R}\oplus\mathbb{C}^3$, $s^2 + 2z^\dagger z = 1$, target metric $\mathrm{d}s^2 + 2\,\mathrm{Re}(\mathrm{d}z^\dagger \mathrm{d}z)$, and $w_0\kappa_\eta > 0$. For $\eta_q = (\cos q, \sin q\, e_1/\sqrt{2})$ with constant q and zero connection, $D\eta = 0$,

$F(0) = 0$, and the target current and stress vanish. For a radial meridian $q(r)$, regularity sets the conserved flux

$$r^2 e^{\nu} \sqrt{f_r}\, w_0(\kappa_\eta + X^3) q' = 0, \qquad X = f_r (q')^2 \tag{A2}$$

to zero. Positivity gives $q' = 0$. This proves the null only for the stated source-free regular boundary problem.

A constant background still permits positive perturbation stiffness. For a nonconstant Dirichlet phase perturbation, the quadratic energy and its second variation (phase self-curvature) are

$$E_\eta^{(2)}[\varphi] = \tfrac{1}{2} w_0 \kappa_\eta \int e^{\Phi - \Psi} |\nabla \varphi|^2 \, \mathrm{d}^3 x,$$

$$\delta^2 E_\eta[\varphi, \varphi] = 2 E_\eta^{(2)}[\varphi] > 0.$$

The mixed metric/phase response vanishes at $D\eta = 0$. The solved matter body therefore has ordinary nonzero lensing and zero additional response in this target family. A vanishing matched seam contribution is not a statement that all possible gravitational or boundary sources vanish. The rolling scalar T in Equation (4) is a different explicitly specified field.

Appendix B. Moving Measured-Redshift Endpoints

For an observer-normalized affine ray, $J_a'' = \mathcal{T} J_a$ with $(J_a, J_a')_a = (0, \omega_a I)$. Let $Z_{\lambda,a} = \delta \ln \omega|_{\lambda_a}$ and $Z_{\chi,a} = \delta \ln \omega|_{\chi_a}$. On the reference ray,

$$\begin{aligned} Z_{\chi,a} &= Z_{\lambda,a} - \mathcal{H}_a \delta\chi_{\mathrm{ray},a}|_\lambda, \\ \delta\lambda_a &= -\frac{Z_{\lambda,a}}{(\ln \overline{\omega})'_{\lambda,a}}, \qquad \delta\chi_a = \delta\chi_{\mathrm{ray},a}|_\lambda + \overline{\chi}'_{\lambda,a} \delta\lambda_a = -\frac{Z_{\chi,a}}{\mathcal{H}_a}. \end{aligned} \tag{A3}$$

Thus converting an affine shift alone does not give the full radial event shift. The moving-vertex initial conditions are

$$\delta J_a|_a = -\omega_a I \, \delta\lambda_a, \qquad \delta J_a'|_a = \Delta\omega_a I, \qquad \Delta\omega_a = \delta\omega_a|_\lambda + \overline{\omega}'_a \delta\lambda_a. \tag{A4}$$

At fixed measured redshift and observer normalization, $\Delta\omega_a = 0$. Output event and frame conversions must also be included. These conventions are needed for the three compatible maps; they are inactive in the local comparison where the maps are held fixed.

Appendix C. Leading Scalar Principal Coefficients

After eliminating the leading metric mixing, the homogeneous time coefficient is

$$D = 1 + \frac{12Hv}{\lambda} + \frac{6v^4}{M_{\mathrm{Pl}}^2 \lambda^2} = 4.3962210287. \tag{A5}$$

For $\mathcal{S} = \sqrt{C^2 + 16 b_T G M(<r)/(\lambda r^3)}$, the leading local diagonal coefficients are

$$K_r = \mathcal{S}, \qquad K_\Omega = \mathcal{S} + \frac{1}{2} r \mathcal{S}', \qquad K_t = D + \frac{3}{2}(\mathcal{S} - C) + \frac{1}{2} r \mathcal{S}'. \tag{A6}$$

Nonnegative density implies $K_\Omega \geq (\mathcal{S}^2 + 3C^2)/(4\mathcal{S}) > 0$. For this branch K_t is also positive; the frozen homogeneous ratio is $C/D = 0.9642051$. The exterior maximum of K_r/K_t is 1.47366 at approximately 100.107 kpc. These are leading local coefficients. Mixed time–space and higher weak-field terms, the full coupled spectrum, and an ultraviolet completion are outside this check.

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